Orthogonal risks in 21st-century sea-level rise: independent roles for emissions reduction and ice-sheet stabilization

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Abstract

Sea-level projections are conventionally presented as functions of emission scenario, implying that emissions reduction is the primary lever for managing coastal risk. We show that this framing obscures the dominant source of uncertainty. Using a data-driven Bayesian forecast calibrated against independent observations of each sea-level contributor, we project 0.98–1.35 m of global mean sea-level rise by 2100 (medians across policy-relevant pathways relative to pre-industrial), exceeding IPCC AR6 estimates by 72–81%. A variance decomposition reveals that 92% of total forecast uncertainty originates from a single source—the dynamical response of the West Antarctic Ice Sheet (WAIS)—while emission-pathway differences account for only 8%. These two dimensions of risk are approximately orthogonal: emissions reduction compresses the predictable, temperature-sensitive components (ocean thermal expansion, glaciers, Greenland) but leaves the WAIS-dominated tail largely unchanged, while interventions targeting ice-sheet stability would collapse the dominant uncertainty without requiring further emissions cuts. Even under aggressive mitigation (2 °C trajectory), we find a 6% probability of exceeding 2 m; under 5 °C warming, this rises to only 19%, confirming that the tail risk is largely emission-insensitive. These results reframe sea-level risk management as a two-lever problem: emissions reduction to shift the central estimate, and ice-sheet research and intervention to reduce the catastrophic tail. Neither alone is sufficient; both are necessary.

Rates of global mean sea-level rise (GMSLR) have doubled twice in the past 50 years [1, 2] (Fig. 1). The first doubling took about 16 years and 0.34 °C of warming [3]. The second took about 21 years and 0.41 °C of warming (Fig. 1). The rate could double again around 2080 if the current observed sea-level acceleration continues, and by 2040 if it takes no more warming than the previous two doublings. The 40-year gap illustrates the uncertainty that coastal planning and adaptation must absorb.

Every centimeter of sea-level rise exposes roughly one million additional people to annual coastal flooding [6], approximately 90% of whom live in low- and middle-income countries [7]. Without additional adaptation, annual flood damages are projected to reach US$14–18 trillion per year for 0.9–1.1 m of rise [8]. Estimates of total adaptation costs per unit sea level rise are not available but a lower-bound, rough-order-of-magnitude estimate of US$4 trillion per meter of sea-level rise is reasonable based on a projected cost of US$400 billion simply to raise dikes by 1 meter along only 3.4% of coastlines [9]. Developing countries currently receive US$21 billion per year in total adaptation finance across all sectors, not just coastal [10], raising serious questions about the feasibility of coastal adaptation at relevant scales and highlighting the extreme financial costs, to say nothing of human and ecological impacts, arising from large uncertainties.

The rate of reduction in the uncertainties is a crucial consideration because effective planning requires reliable sea-level forecasts be delivered before key decisions need to be made. Uncertainty in the rates of sea-level rise compounds uncertainty in inundation thresholds and paralyzes anticipatory spatial planning, routinely defaulting policy toward *in situ* maladaptations that compound sunk costs and strand assets [11], or delaying action until forced by disaster [12]. Unconstrained timelines encourage fragmented, market-based incrementalism that deploys public capital inefficiently [13] and systemically transfers the financial burden of reactive retreat onto vulnerable municipalities and demographics [14].

Current coastal adaptation planning relies heavily on projections from the IPCC Sixth Assessment Report (AR6), which estimates 0.4–0.7 m of rise by 2100 relative to the 1995–2014 baseline [15, 16]. These projections depend largely on process-based ice sheet models that diverge from satellite observations within years of initialization [17, 18]. As a result, observed GMSL trends (0.66 m median at 2100, 0.41–0.91 m 90% CI) already exceed AR6 medium-confidence projections under all policy-relevant emission scenarios [2, 19]. AR6 itself documents this in a single figure but without comment: extrapolating the satellite-era rate and acceleration—which reflect the current warming trajectory that could lead to 3 °C (SSP2-4.5)—produces sea-level rise that matches or slightly exceeds the process-based projections under the very high emissions scenario (SSP5-8.5), which is now regarded as generating an implausible level of warming (AR6 Figure 9.27; [15, 20]). So while the data show that rates of sea level rise double with approximately 0.4 °C of warming (Fig. 1b), IPCC models need roughly an order of magnitude more warming just to reproduce current observed rates of sea-level rise.

Disagreement between observations and ice sheet models arise from known structural biases, including the absence of leading-order physical processes like iceberg calving [18] and an assumed ice viscosity that is less sensitive to stress than observations indicate [21]. The latter produces a roughly 30% underestimate of projected 21st century ice loss, exceeding the uncertainty due to warming trajectories [22]. More broadly, decades of general forecasting research shows that models more complex than their calibration data can constrain produce worse predictions than simpler models [23, 24]. Current ice sheet models have millions of degrees of freedom representing spatially distributed parameters and uncertain parameterizations. The observations used to initialize these models inform only a small fraction of these degrees of freedom, with the remainder determined by prior assumptions and model design [25, 26]. A forecasting system based on models representing diverse approaches and a hierarchy of complexity is needed to fill the needs of coastal planning for sea-level rise.

Data-driven forecasting

We construct a data-driven forecast that focuses on different sea-level contributors, each with minimal physical models constrained by independent observational records spanning decades. Observed GMSLR and its sensitivity to observed global mean surface temperature (GMST) are diagnostics, and we show that the sum of our model components agrees within

6% of 2000–2025 observed GMSL in hindcasting tests (Supplementary Material). Unless otherwise noted, all sea-level values in the main text are global-mean 21st-century contributions referenced to the year 2000, while values in the abstract are relative to pre-industrial for consistency with existing projections and literature. Throughout, we label emission scenarios by their approximate end-of-century global mean surface temperature: 2 °C (SSP1-2.6), 3 °C (SSP2-4.5), and 4 °C (SSP3-7.0).

Our framework builds on two established lines of data-driven forecasting. The simplest is the "naive" forecast, which requires only a fit to the observed GMSL rate and acceleration. The satellite altimetry record provides GMSLR measurements from 1993–2025 that show a constant acceleration of 0.08 $\mathrm{mm\,yr^{-2}}$ (0.02–0.14 $\mathrm{mm\,yr^{-2}}$) and rate 4.5 $\mathrm{mm\,yr^{-1}}$ (3.5–5.5 $\mathrm{mm\,yr^{-1}}$) in 2025 [2, 19]. Extrapolating these values to 2100 yields 66 cm (41–91 cm) of 21st-century rise (Fig. 3a). Assuming the naive forecast has predictive skill over about half the calibration window, the extrapolation of the 32-year satellite record provides predictive skill over the next 15-20 years. Beyond that horizon, the naive forecast is less reliable but nevertheless establishes a scenario-independent, data-driven reference (Fig. 3a).

At the next level of complexity are the semi-empirical models that are based on the observed sensitivity of the rate of GMSLR to changes in GMST (Fig. 1b). Under current warming trends, Rahmstorf (2007) [5] predicts 0.67 m (0.63–0.71 m) of 21st-century rise, matching the extrapolation of modern observed rates within error [2]. Although physically motivated by the fit to observed sensitivities, semi-empirical models were assessed as *low confidence* by IPCC AR5 because they do not resolve individual contributing processes and because extrapolation beyond the calibration period assumes a stationary temperature–sea-level relationship [27]. This reasoning is carried forward by AR6 but violated by the decision to favor process models that omit key physical processes, are calibrated on non-stationary data, and were not tested against observed GMSL, the very quantity they claim to predict, prior to publication [15]. Our own results show that the simplest semi-empirical models have much greater forecasting skill for the components of sea-level rise that are directly sensitive to warming.

Constructing our component model

We decompose global mean sea level into separate components: ocean thermal expansion (thermosteric), Greenland Ice Sheet (GrIS) surface mass balance (SMB) and glacier discharge, West Antarctic Ice Sheet (WAIS) discharge, East Antarctic Ice Sheet (EAIS) SMB, Antarctic Peninsula total ice loss, glaciers (land ice not in the ice sheets), and terrestrial water storage. Our component-summation approach overlaps with the methodology of previous studies [28–30], while extending to more recent data, rate-space blending against satellite-era observations to ensure alignment between the forecast and observed data trends (Supplementary Materials), and a structured deep-uncertainty treatment of WAIS discharge. The sensitivities of glaciers, GrIS discharge, and Antarctic Peninsula to warming are calibrated with independent observational datasets spanning 24–71 years (Table S1). Surface mass balance for GrIS and EAIS are adopted from published model sensitivities rather than data calibration because they lack sufficient data coverage for our approach [31, 32]. For completeness, we adopt the IPCC projection of terrestrial water storage due to a lack of suitable data to support our approach. Details of our models for EAIS and Antarctic Peninsula can be found in Supplementary Materials.

The thermosteric component is the largest single contributor to observed sea-level rise. We model it with a two-layer energy balance [38] fitted jointly to NOAA *in situ* thermosteric measurements at 0–700 m (1957–2025) and 0–2000 m (2005–2025) [39] (Fig. 2a). In our model, the upper ocean responds to GMST warming through a thermal state variable S_u with relaxation time τ_u. The deep ocean thermal state relaxes toward S_u on a longer timescale $\tau_d = 150\,\mathrm{yr}$ [38]. We fit the model parameters to the data using a Monte Carlo approach described in Supplementary Material. Forecasts are generated by driving the calibrated two-layer model with IPCC AR6 temperature trajectories, propagating posterior parameter uncertainty via Monte Carlo sampling (Supplementary Material).

The Greenland Ice Sheet is currently the largest land-ice contributor. Greenland's contribution is to imbalances between mass gained through snow accumulation and mass loss from melting and glacier ice discharge. We treat SMB and ice discharge separately. We found that available data are insufficient to constrain SMB. Instead, we use regional climate models. For the hindcast, we use RACMO-derived SMB observations [33, 40] directly. For projections, we adopt published linear and quadratic sensitivities of SMB to GMST from RCM intercomparisons [31, 32], using the spread across RCMs (MAR, RACMO, HIRHAM) as structural uncertainty [41]. The SMB rate anomaly is then integrated forward under each emissions pathway (Supplementary Material).

GrIS discharge acceleration is sensitive to ocean temperature [42], and we model it as the cumulative response to time-delayed subsurface ocean warming, with a sensitivity γ, a delay δ between ocean warming and glacier response, and a secular drift r_0 (Supplementary Material). We force the model EN4 ocean temperatures [43] averaged over 200-500 m depth in the peripheral waters around Greenland. We cross-correlate the ocean temperatures against glacier discharge [33] to find the delay to be $\delta = 5$ yr (Fig. 2b), consistent with fjord overturning and calving dynamics (Supplementary Material). The fitted sensitivity $\gamma = 0.37\ \mathrm{mm\,yr^{-1}\,{}^{\circ}C^{-1}}$ is consistent with physical scaling estimates from ocean and glacier dynamics ($0.36\ \mathrm{mm\,yr^{-1}\,{}^{\circ}C^{-1}}$), suggesting that our 3-parameter model is capturing the salient physical processes governing the acceleration of Greenland glaciers over the entire 54-year observational record. An independently derived discharge record (1986–2023) [35] and NOAA Arctic Report Card estimates [34], both withheld from calibration, fall within the 90% credible interval, as does the total Greenland mass balance (Fig. 2c).

Mountain glacier mass loss is modeled as a rate–temperature relationship calibrated against the GlaMBIE global consensus [2000–2023; 44] and GMST. The data only support a linear sensitivity (Supplementary Material), which we find to be $b_{\mathrm{gl}} = 0.66\ \mathrm{mm\,yr^{-1}\,{}^{\circ}C^{-1}}$ (Fig. 2d). This is approximately five times the value embedded in IPCC AR6 median projections ($\sim$0.13 $\mathrm{mm\,yr^{-1}\,{}^{\circ}C^{-1}}$) and 1.3–1.7$\times$ the effective sensitivity of the PyGEM process model [45], but is overall a small contributor to sea-level rise (Supplementary Material).

We compare the sum of the independently calibrated components to observations of GMSL from satellite altimetry as a diagnostic (Fig. 2e,f). The diagnostic is possible because each component is calibrated against its own observational record. Total GMSL is not used as a calibration target. The component sum closes 93–101% of the observed GMSLR rate over three epochs (1993–2025, 2000–2025, 2010–2025), with a variance-weighted attribution assigning the residual mainly to the thermosteric (62%) and terrestrial water storage (23%) components, both within their stated uncertainties, and all components are within 90% CI of their independent estimates (Supplementary Material). The transient sensitivity of our component model is also in close agreement with the observed GMSLR rate (Fig. 1b) and the Rahmstorf (2007) [5] semi-empirical results.

West Antarctic Ice Sheet.

WAIS is the largest source of uncertainty in sea-level projections (Figs. 2g,h). The ice sheet contains ∼3.3 m of sea level equivalent and rests on land that is below sea level and deepens inland, making it susceptible to internal feedbacks commonly called the marine ice sheet instability (MISI; [46]). The delivery of warm, deep seawater is the primary climate driver of mass loss [18]. Ocean thermal forcing sufficient to drive retreat is likely committed throughout the 21st century regardless of emissions pathway [47, 48] and may be driven as much by oceanographic processes local to the ice sheet as nonlocal processes [49]. Based on these lines of evidence, we make our WAIS projections independent of the warming trajectory, meaning the range of 21st-century outcomes is controlled by ice dynamics that, to first order, are insensitive to emissions reductions over our forecasting horizon.

No physical process models have been proven to produce reliable projections of WAIS mass loss [17] and we find insufficient data coverage and duration to support a reliable data-driven forecast like those described above. To provide a representative range of uncertainties, we construct a scenario-mixture framework with three outcomes that are grounded in the scientific literature (Fig. 2g,h; Supplementary Material). Scenario 1: 'slow WAIS' where rates of discharge continue to increase within the range of present-day rates, contributing 25–85 mm GMSLR by 2100. Based on the changes in WAIS discharge over the past decades and the position by the IPCC AR6, we take this scenario to be unlikely, and assign it a 10% probability. Scenario 2 represents the role of MISI-driven acceleration with grounding-line retreat on retrograde bedrock, with a GMSL contribution of 150–1000 mm by 2100. The lower bound is anchored to transient-calibrated ice sheet model projections [50] and the observed 33 km grounding-line retreat at Pine Island Glacier since 1992 [51] while the upper bound is set by structured expert judgment [52]. We assign Scenario 2 an 80% probability. Scenario 3 represents MISI combined with marine ice cliff instability and its values defined by the IPCC AR6 low-confidence 83rd (600 mm) and 95th (1300 mm) percentile ranges. We assign a 10% probability to this scenario. Within each scenario, the 2100 endpoint is drawn from a skew-normal distribution in log-space with physically motivated asymmetry [53]. A multiplicative rheology correction (R with median $R = 1.3$) is applied to all scenarios, reflecting the underestimation of ice loss when models assume the ice-viscosity stress exponent $n = 3$, which is contrary to the observationally constrained value $n = 4.1 \pm 0.4$ [21, 22, 54]. The resulting mixture has a median 21st century sea-level rise of 0.41 m with a 0.07 to 1.56 m 90% CI at 2100. We refer to this mixed model as 'fast WAIS' because it allows for, but does not force, rapid ice mass loss (Scenarios 2 and 3). Our choices of scenario mixture weights have small quantitative impacts on our forecast results in the median and high-end tails of the distribution, but do not qualitatively impact the findings of this study (Supplementary Materials).

Forecasts

We focus on warming trajectories of 2 °C (SSP1-2.6) through 4 °C (SSP3-7.0), the pathways most consistent with current policies and stated pledges (Fig. 3). Under the 2 °C target, our forecast shows 0.79 m of 21st-century sea-level rise [0.46–1.89 m], which is 81% above the IPCC AR6 medium-confidence estimate of 0.44 m. Under 3 °C warming, approximately the current trajectory, the median is 0.96 m [0.62–2.07 m], 72% above IPCC's 0.56 m. For the 4 °C trajectory, the median reaches 1.16 m [0.82–2.26 m]. For reference, the 5 °C warming trajectory (SSP5-8.5) yields a median of 1.35 m [0.99–2.45 m]. All sea-level values given relative to year 2000.

Relative to pre-industrial GMSL (adding ∼0.19 m of observed rise through 2000), these forecasts correspond to medians of 0.98–1.35 m of global mean sea level rise across the policy-relevant pathways. The probability of exceeding 1 m relative to pre-industrial by 2100 is 47% under 2 °C warming, 76% under 3 °C, and 96% under 4 °C warming trajectory. Averaged across these pathways, the probability of exceeding 1 m of sea level rise relative to pre-industrial levels is 73%, and is largely insensitive to the relative weighting of scenarios (Fig. 3) because forecasting spread is driven by the dynamics of WAIS glaciers. If, however, the 'slow WAIS' scenario were to be true, the probability of exceeding 1 m of sea level rise relative to pre-industrial levels drops to zero if warming stays below 3 °C (Fig. 3b).

Orthogonal dimensions of sea-level risk

The stark differences between the probability of exceeding 1 m of (pre-industrial) sea-level rise between a slowly evolving WAIS and a rapidly evolving WAIS highlights the dominant role of WAIS in sea-level rise projections.

The within-scenario 90% uncertainty range (1.47 m for 3 °C) is four times larger than the spread across scenario medians (0.37 m from 2–4 °C). A variance decomposition (law of total variance across policy scenarios) attributes 92% of total forecast variance to within-scenario uncertainty—predominantly WAIS—and 8% to between-scenario differences. This is not merely a statement about the relative magnitude of uncertainties. It reveals that sea-level risk decomposes into two approximately orthogonal dimensions, each addressed by a distinct class of intervention.

The first dimension is *emission-sensitive*. The temperature-dependent components—ocean thermal expansion, glacier mass loss, Greenland surface mass balance and glacier discharge, and the Antarctic Peninsula—respond to global temperature through well-characterized physical relationships calibrated against decades of observations. Their collective contribution shifts systematically across warming trajectories and is directly reducible by emissions cuts. Conditional on a stable WAIS, the forecast narrows to a 90% range of approximately 140 mm, a level of precision comparable to the thermosteric projection alone (Fig. 3a, purple band). In this regime, emissions policy is the primary lever: the difference between 2 °C and 4 °C warming corresponds to roughly 200 mm of additional rise, nearly all of which is attributable to temperature-sensitive components.

The second dimension is *emission-insensitive*. WAIS dynamics accounts for 91% of total forecast variance and is largely decoupled from the emission pathway because ocean thermal forcing sufficient to sustain grounding-line retreat is already committed [47, 48]. The tail of the forecast distribution—the outcomes that drive the most severe human and economic consequences—is controlled almost entirely by whether WAIS retreat remains gradual or transitions to rapid, irreversible collapse. Emissions reduction compresses the median but barely touches this tail: the probability of exceeding 2 m relative to pre-industrial is 6% under the 2 °C trajectory and 19% under 5 °C. This factor-of-three difference is meaningful, but it does not eliminate the risk. Under any plausible emission pathway, there remains a non-negligible probability of catastrophic sea-level outcomes that emissions reduction alone cannot address.

The orthogonality of these two risk dimensions has a direct implication for resource allocation. Emissions reduction and ice-sheet stabilization are not competing priorities—they are complementary interventions operating on independent components of the problem. Emissions reduction without ice-sheet stability yields a well-constrained forecast centered on a potentially catastrophic outcome. Ice-sheet stability without emissions reduction yields a safe ice-sheet contribution but a wider envelope from thermally driven components. Adequate risk management requires both.

Implications for ice-sheet intervention

The concentration of forecast uncertainty in a single physical system—and specifically in a single binary question about the reversibility of WAIS retreat—points to the value of research and potential interventions aimed at ice-sheet stabilization. Proposed interventions, including submarine barriers to restrict warm water access to ice-sheet grounding zones [?], targeted cooling of critical ice shelves, and grounding-line stabilization through sediment deposition, remain at an early conceptual stage. Our results provide quantitative motivation for advancing this research: if any intervention could shift probability mass from the unstable to the stable WAIS regime, the expected reduction in sea-level rise—and its associated human and economic costs—would be large. The stable-WAIS forecast (90% range of 540–680 mm) versus the full forecast (90% range of 600–2100 mm) represents a reduction in upper-bound risk of approximately 1.4 m, corresponding to hundreds of millions of fewer people exposed to coastal flooding [6] and trillions of dollars in avoided damages [8].

Furthermore, the forecast uncertainty is not smoothly distributed across outcomes but is concentrated in two distinct regimes. Conditional on a stable WAIS, the 90% range is approximately 140 mm. Conditional on ice-sheet instability, it expands to approximately 1500 mm with a median nearly 400 mm higher. The unconditional forecast—a probability-weighted mixture of these regimes—produces a wide confidence interval that reflects the superposition of two narrower distributions rather than a single diffuse uncertainty. A coastal community designing for the unconditional median prepares for an outcome that falls between the two most probable futures, potentially underbuilding for instability while overbuilding for stability. Adaptation strategies should therefore be designed in stages: sized for the stable regime now, with identified triggers and pre-planned expansions if observations confirm the onset of instability.

Reducing this structural uncertainty requires resolving a specific physical question: whether the observed acceleration in West Antarctic mass loss reflects a transient response to ocean forcing or the early stages of irreversible retreat. This is, by a wide margin, the single most consequential open question in sea-level science, and the one where targeted investment in observations, process understanding, and intervention research would yield the greatest return in forecast precision and risk reduction.

Broader implications

The data-driven framework presented here is not a replacement for process models, which remain essential for understanding the mechanisms of ice-sheet change. Our framework is complementary: an independent, data-driven estimate that makes its assumptions and limitations explicit, provides an observational benchmark against which process models can be tested, and produces conditional distributions that reflect what the data currently support. We consider robust, well-calibrated process models to be high-value targets for research and development, but the evidence suggests these models are not ready to be used for planning that costs billions of dollars and impacts millions of lives.

We emphasize that the disconnect between the high costs of sea-level rise and the large uncertainties in current sea-level projections is a consequence of chronic under-investment compounded by the lack of institutional homes able to focus on translating knowledge gained through basic scientific research into forecasts for coastal communities in the way that, for example, the National Weather Service has traditionally done [55]. A specific example is that modeling efforts for ISMIP6, which informed IPCC AR6 projections, were largely unfunded, leading to a dearth of model runs and near-absence of model parameter-sensitivity studies [17]. The world is going to endure high costs from sea-level rise (Fig. 3b). The questions are: how high will those costs be, and will we invest now to understand where and how to target funding to reduce the harm to people, ecosystems, and infrastructure, or continue on our current path? Our analysis shows that this investment must be directed along two independent axes—reducing emissions to compress the predictable components, and understanding and potentially stabilizing the West Antarctic Ice Sheet to address the dominant, emission-insensitive uncertainty that defines the upper bound of 21st-century sea-level rise.

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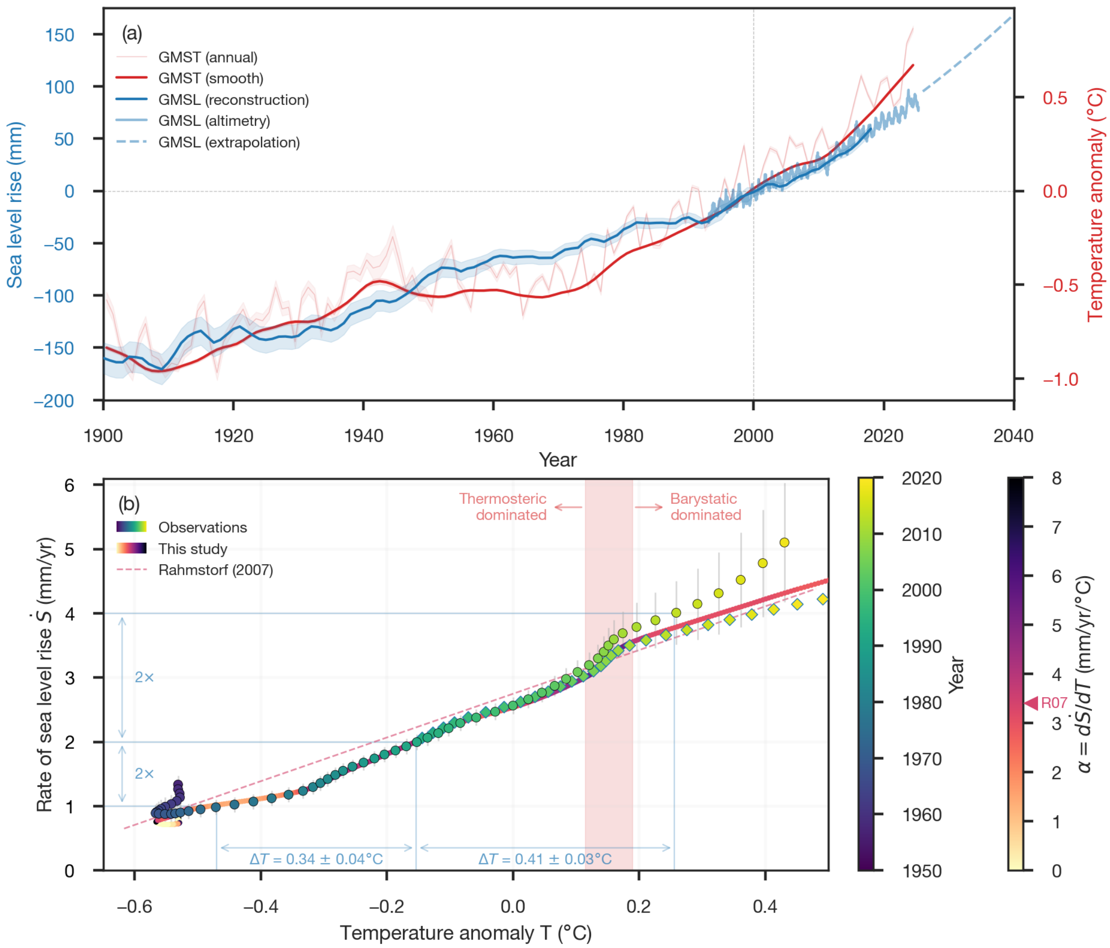

Figure 1: **Observed global mean sea-level rise and its response to global mean surface temperature. (a)** GMSLR (blue, 5-year smooth; [1]) and NASA satellite altimetry (light blue; [4]), with the satellite-era extrapolation (dashed; [2]). GMST (red, 15-year smooth [3]) is referenced to the righthand axis. **(b)** Rate of GMSLR ($\dot{S} = dGMSLR/dt$) versus GMST anomaly, T. Observed values are shown in circles [1] and diamonds [4], colored by year, and the component model of this study is a line colored by instantaneous transient sensitivity $\alpha = d\dot{S}/dT$. The Rahmstorf (2007) semi-empirical model is shown in a color corresponding to its α value [5]. Horizontal and vertical blue lines mark the 1–2 and 2–4 mm yr^{-1} rate doublings that occurred during 1976–1992 (1959–1993 with $\pm 1\sigma$) and 1992–2013 (1992–2014, $\pm 1\sigma$), respectively. The corresponding warming increments (ΔT) show that the first doubling required less warming (0.34 ± 0.04 °C) than the second (0.41 ± 0.03 °C). The vertical red band indicates the hand off from thermosteric (ocean thermal expansion) being the primary contributor to sea-level rise to barystatic (changes in glacier ice and water storage) being the dominant contributor. The sinuous tail in the early observations is largely due to dam impoundment of water rather than a thermodynamic signal [1].

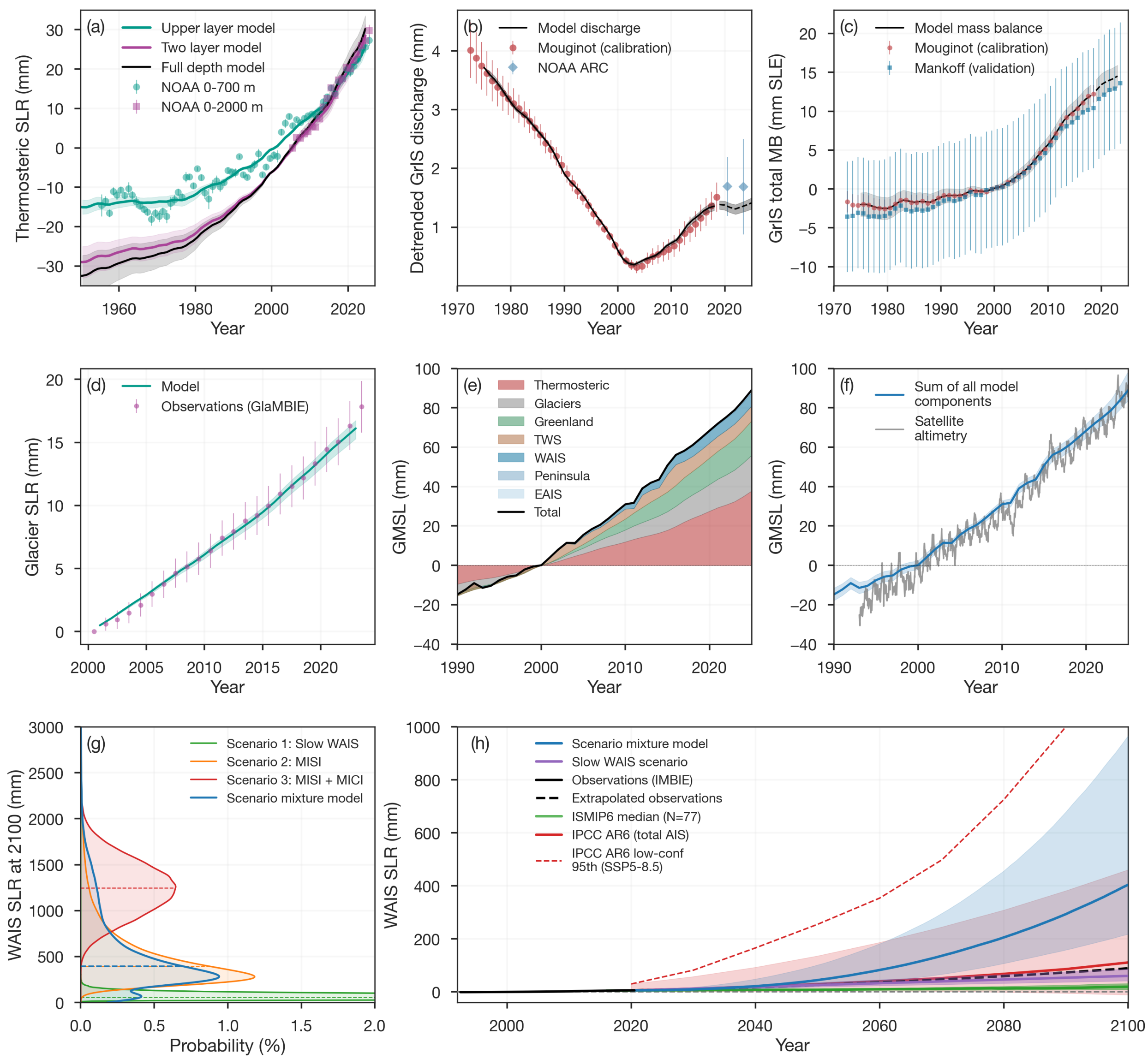

Figure 2: **Model calibration, validation, and WAIS uncertainty.** **(a)** Our two-layer energy balance model fitted jointly to NOAA thermosteric values at different depth layers with the full depth model used in the forecasts (Supplementary Materials). **(b)** Greenland ice discharge modeled as a cumulative time-delay response to subsurface ocean warming with the calibration data from [33] shown with filled circles and diamonds showing NOAA Arctic Report Card estimates [34]. **(c)** Greenland total mass balance with validation data shown in blue squares with 90% CI error bars [35]. **(d)** Global glacier model fitted to GlaMBIE consensus observations (2000–2023) using a linear temperature-driven model. **(e, f)** Budget closure: temperature-forced component hindcast (stacked) compared with satellite altimetry data. **(g)** WAIS scenario distributions at year 2100. **(h)**WAIS projections from our full scenario mixture (blue) and slow WAIS scenario (purple) compared with observations (black, [36]), the ISMIP6 ensemble (green; [37]), and IPCC AR6 projections (red).

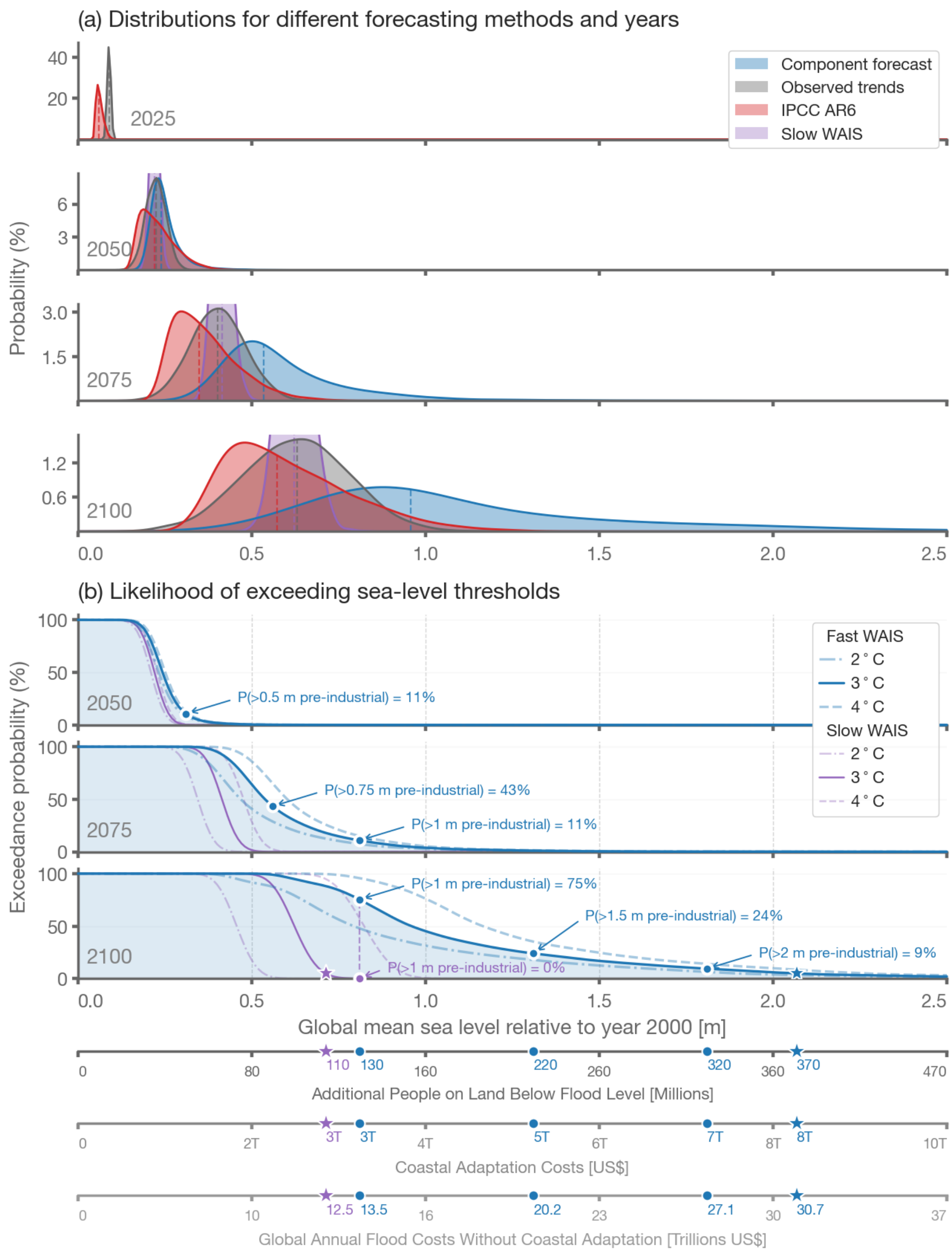

Figure 3: **Probability distributions for global mean sea-level projections.** (a) Probability distribution for the composite forecast of this study with the WAIS scenario mixture model (blue), the composite forecast of this study with the 'slow WAIS' scenario (purple, clipped for clarity), extrapolation of observations (grey), and IPCC AR6 projections (red). Vertical dashed lines indicate the respective medians. (b) Probability of exceeding a given sea-level threshold for the 2°C (dash-dot), 3°C (solid), and 4°C (dashed) warming trajectories including the WAIS scenario mixture model (blue) and slow WAIS scenario (purple). Lower axes show estimates of the corresponding number of people currently living on land that will be below flood level [6], global annual coastal adaptation costs (adapted from [9] assuming the total adaptation cost is only 10x the cost of raising dikes along 3.4% of global coastlines), and global annual flood costs assuming no additional coastal adaptation [8]. Stars indicate the 5% exceedance probability for year 2100.

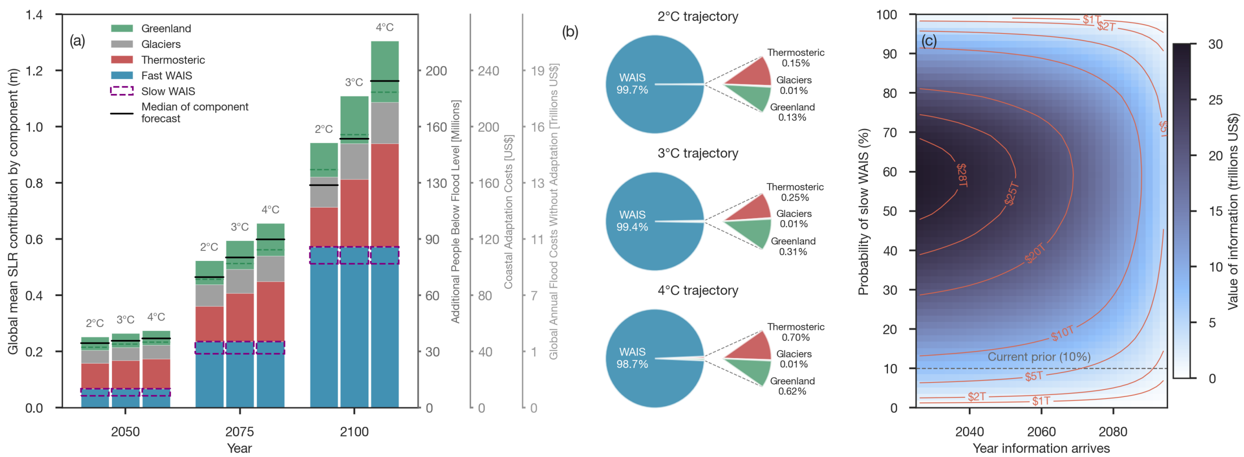

Figure 4: **Component contributions, variance decomposition, and value of information.** **(a)** Mean sea-level contribution by component at 2050, 2075, and 2100 for 2 °C, 3 °C, and 4 °C warming trajectories. WAIS (bottom, blue) is emission scenario-invariant. All other components are emission-sensitive, and the value of emissions reduction can be traced by the stack heights and median values (black lines). Dashed purple boxes show the slow WAIS only scenario. Green dashed lines within Greenland bars partition discharge anomaly (below) from SMB anomaly (above). Gains in EAIS approximately offset losses in Peninsula, so we omit both. Right-hand axes show corresponding societal impacts as in Fig. 3. **(b)** Variance-fraction pie charts at 2100 for each warming trajectory. WAIS accounts for 97–99% of total forecast variance. Fan breakouts show the non-WAIS contributions. **(c)** Financial value of resolving WAIS uncertainty as a function of when information arrives and the probability of slow WAIS decline. Dashed line marks the current prior on WAIS Scenario 1 (10%). Contours and colormap in trillions US$ based on coastal adaptation costs (Supplemental Material).